

ImageNet Challenging Classification with the Raspberry Pis: A Federated Learning Algorithm of Local Stochastic Gradient Descent Models

Thanh-Nghi Do^{1,2}, Minh-Thu Tran-Nguyen^{1,2}

¹ College of Information Technology
Can Tho University, 92000-Cantho, Vietnam

² UMI UMMISCO 209 (IRD/UPMC)
Sorbonne University, Pierre and Marie Curie University - Paris 6, France
{dtngchi, tnmthu}@ctu.edu.vn

Abstract. In this paper, we propose the federated learning algorithm of local stochastic gradient descent (SGD) models at edge devices (i.e. Raspberry Pis) to classify large ImageNet dataset having 1,281,167 images with 1,000 classes. The full very large training dataset is divided into subsets which are stored in local Raspberry Pis. And then, the federated learning algorithm uses Raspberry Pis to train in the incremental and parallel way local SGD models from their own subset without exchanging data. The incremental local SGD tailored on Raspberry Pi sequentially loads small data blocks of its own local training subset to learn local SGD models. In which, the local SGD algorithm uses k means to split the data block into k partitions and then it learns in the parallel way SGD models in each data partition to classify the data locally. The numerical test results on Imagenet dataset show that our federated learning algorithm of local SGD models with 4 Raspberry Pis (Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz, 4GB RAM) is faster than the state-of-the-art linear SVM run on a PC (Intel(R) Core i7-4790 CPU, 3.6 GHz, 4 cores, 32GB RAM) with the competitive classification accuracy.

Keywords: ImageNet classification · Federated learning · Incremental local SGD · Raspberry Pi.

1 Introduction

The most challenge in the computer vision community is the classification of ImageNet dataset [5], [6] with more than 14 million images for 21,841 classes, due to the large scale number of images and classes. The efficient image classification algorithms allow to find what we are looking for in very large amount of images produced by internet users. The image classification task automatically categorizes the image into one of predefined classes. It consists of two key stages: the feature extraction and the machine learning scheme.

The classical approaches [1], [7], [9], [12], [13], [25], [37], [33], [42] proposed to use popular handcrafted features such as the scale-invariant feature transform

(SIFT [26], [27]), the bag-of-words model (BoW) and then to train Support Vector Machines (SVM [40]) to classify images. Recent convolutional neural networks (CNN [24]), deep neural networks including VGG19 [36], ResNet50 [17], Inception v3 [38], Xception [4], EfficientNet [39] aim to learn visual features from images and the classifier in an unified algorithm to efficiently classify images. These deep networks achieve the prediction correctness more over 70% for ImageNet challenging dataset.

For handling the classification task of ImageNet challenge, we propose to use the pre-trained deep learning network EfficientNet [39] without the top layer (i.e. softmax), to extract invariant features from images of ImageNet dataset. After that, we develop the new federated learning algorithm of local SGD models at the Raspberry Pis to classify ImageNet dataset. Firstly, the full very large training ImageNet dataset is divided into subsets which are stored in local Raspberry Pis. And then, the federated learning algorithm of local SGD models uses Raspberry Pis to train in the incremental and parallel way local SGD models from their own subset without exchanging data. To overcome the main memory limit, the incremental local SGD tailored on the Raspberry Pi sequentially loads small data blocks of local training subset to learn local SGD models. In which, the local SGD algorithm uses k means [28] to split the training data block into k data partitions and then it learns in the parallel way SGD models in each data partition to classify the data locally. The numerical test results on ImageNet dataset show that our federated learning algorithm of local SGD models with 4 Raspberry Pis (Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz, 4GB RAM) is faster than the state-of-the-art linear SVM such as LIBLINEAR [14] run on a PC (Intel(R) Core i7-4790 CPU, 3.6 GHz, 4 cores and 32 GB RAM) with the competitive classification accuracy. The federated learning algorithm of local SGD models on 4 Raspberry Pis takes 18 minutes to classify ImageNet dataset having 1,281,167 images in 1280 deep features into 1,000 classes with an accuracy of 88.07%.

The remainder of this paper is organized as follows. Section 2 briefly presents the federated learning algorithm of local SGD models. Section 3 shows the experimental results before conclusions and future works presented in section 4.

2 Federated learning of local stochastic gradient descent models

Let us consider a classification task with the dataset $D = [X, Y]$ consisting of m datapoints $X = \{x_1, x_2, \dots, x_m\}$ in the n -dimensional input space R^n , having corresponding labels $Y = \{y_1, y_2, \dots, y_m\}$ being $\{cl_1, cl_2, \dots, cl_p\}$.

2.1 Stochastic gradient descent

The stochastic gradient descent (SGD) algorithm tries to find p separating planes for p classes (denoted by normal vectors $w_1, w_2, \dots, w_p \in R^n$) in which the

plane w_i separates the class cl_i from the rest. This is accomplished through the unconstrained problem (1).

$$\min \Psi(w_p, [X, Y]) = \frac{\lambda}{2} \|w_p\|^2 + \frac{1}{m} \sum_{i=1}^m L(w_p, [x_i, y_i]) \quad (1)$$

where the errors are measured by $L(w_p, [x_i, y_i]) = \max\{0, 1 - y_i(w_p \cdot x_i)\}$ and a positive constant λ is to control the regularization strength ($\|w_p\|^2$).

Studies in [2], [35] illustrate that the SGD algorithm solves the unconstrained problem (1) by updating w on T epochs with a learning rate η . For each epoch t , the SGD uses a single datapoint (x_i, y_i) randomly in the mini batch B_i to compute the sub-gradient $\nabla_t \Psi(w_p, [x_i, y_i])$ and update w_p as follows:

$$w_p = w_p - \eta \nabla_t \Psi(w_p, [x_i, y_i]) \quad (2)$$

The SGD is a simple yet efficient algorithm for large-scale learning due to the computational complexity corresponding to $O(mnp)$ (linear in the number of training datapoints m).

In recent last years, it rises powerful, low-cost embedded devices. For example, the Raspberry Pi 4 (Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz, 4GB RAM) is only 55 USD. This leads an increasingly popular choice for machine learning and IoT projects, as illustration in [21], [22], [23], [30], image classification on IoT edge devices [29], MobileNet family tailored for Raspberry Pi [15], running AlexNet on Raspberry Pi [18].

Nevertheless, it is intractable to train the SGD models with the Raspberry Pi for ImageNet challenging problem having 1,281,167 images with 1,000 classes since it requires at least 16 GB RAM for loading the training dataset and the high computational cost.

Our federated learning algorithm of local SGD models at Raspberry Pis aims to reduce the training time and the required main memory for classifying such very large dataset ImageNet.

2.2 Federated learning of local SGD models

The federated learning as illustrated in [20] is to train the decentralized machine learning model (e.g. deep neural networks) across multiple edge devices (e.g. smartphones or Internet of Things - IoT devices), while keeping the training data locally without exchanging data with other devices [16], [19], [43].

Our federated learning algorithm of local SGD models (denoted by FL-*l*SGD) uses Raspberry Pis to train the shared classification model. As showed in Fig. 1, the full training dataset is divided into v subsets which are locally stored in Raspberry Pis. The shared global classification model in the server includes local SGD models which are learned from local training subset at the Raspberry

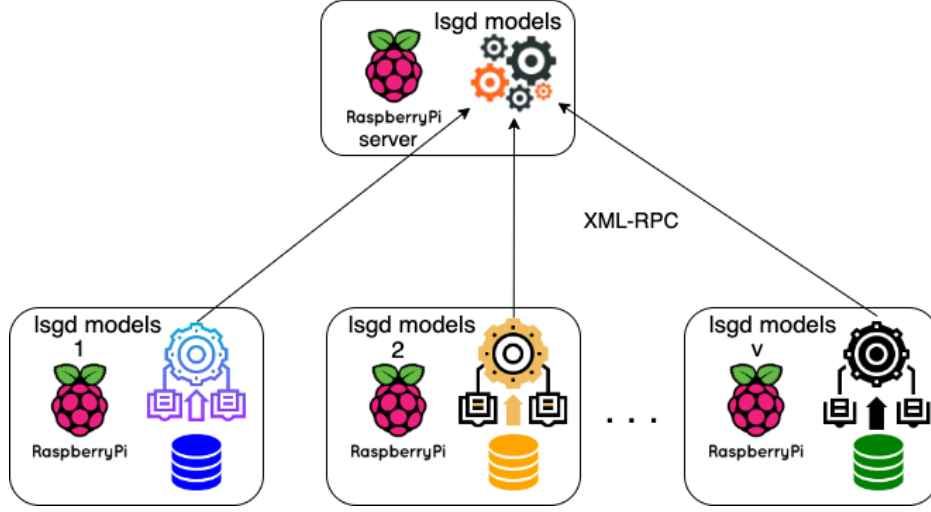


Fig. 1: Federated learning of local SGD

Pi without exchanging training data. The communication used in the federated learning algorithm is XML-RPC protocol¹.

At the Raspberry Pi, our proposed local SGD algorithm (denoted by k SGD in Fig. 2) uses k means [28] to split the training data block into k partitions and then it learns local SGD models in data partitions in parallel way on multi-core computers.

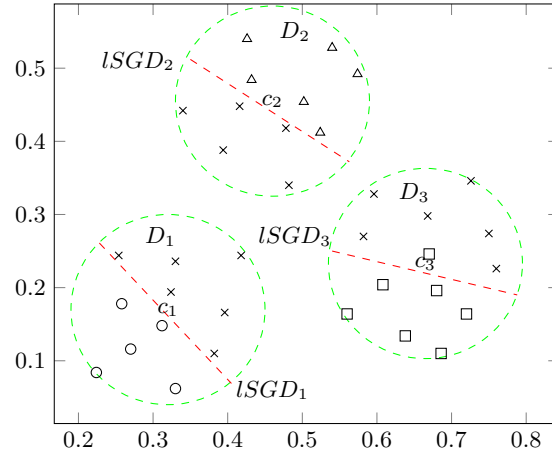
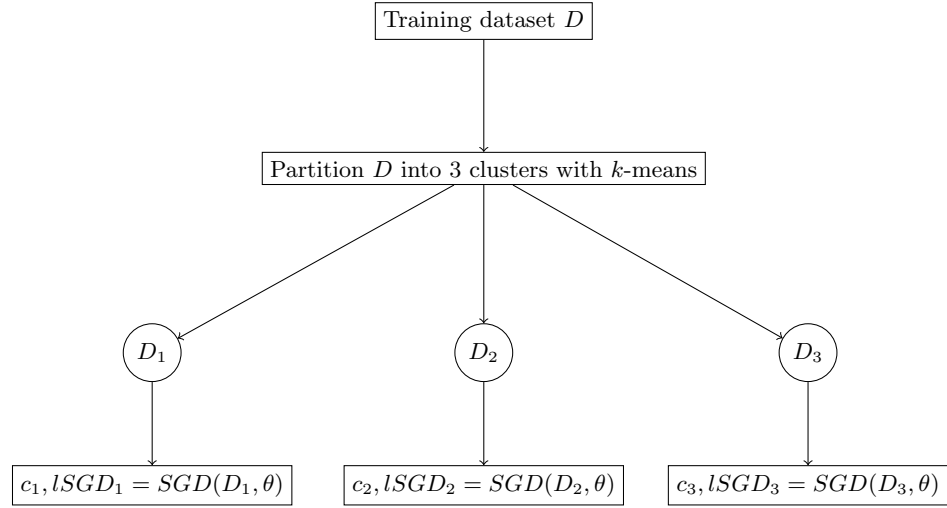
Algorithms 1 and 2 describe the k local SGD learning stage and prediction, respectively. *Code line 5* in Algorithm 1 parallelizes the training task of local SGD models with the shared-memory parallel programming in C on multi-core computers.

The k local SGD algorithm not only reduces the training complexity of the full SGD and but also allows to parallelize the training task of k local SGD models on multi-core computers.

Let us to illustrate the complexity of the k local SGD algorithm. Splitting the full training dataset with m datapoints in n dimensions and p classes into k balanced clusters leads the cluster size being about $\frac{m}{k}$ and the number of classes in a cluster scaling $\frac{p\omega}{k} < p$. The training complexity of a local SGD ² is $O(\frac{m}{k}n\frac{p\omega}{k})$. Therefore, the complexity of parallel training k local SGD models on a P -core processor is $O(\frac{P}{k}\omega np)$. This illustrates that parallel learning k local SGD models is $\frac{Pk}{\omega}$ times faster than the global SGD training ($O(mnp)$).

¹ XML-RPC created in 1998 by Dave Winer of UserLand Software and Microsoft, is a remote procedure call (RPC) protocol which uses XML to encode its calls and HTTP protocol to exchange information between computers.

² It must be noted that the complexity does not include the minibatch k -means [34] used to partition the full dataset.

Fig. 2: Three local SGD model (k SGD)

Algorithm 1: k local SGD algorithm (k SGD)

```

input :
    training dataset  $D$ 
    number of local models  $k$ 
    parameters of SGD  $\theta$ 

output:
     $k$ SGD-model ( $k$  local SGD models)

1 begin
2   /* $k$ -means performs the data clustering on  $D$ ;*/
3   creating  $k$  clusters denoted by  $D_1, D_2, \dots, D_k$  and
4   their corresponding centers  $c_1, c_2, \dots, c_k$ 
5   #pragma omp parallel for
6   for  $i \leftarrow 1$  to  $k$  do
7     /*learning a local SGD model from  $D_i$ ;*/
8      $SGD_i = SGD(D_i, \theta)$ 
9   end
10  return  $k$ SGD-model =  $\{(c_1, SGD_1), (c_2, SGD_2), \dots, (c_k, SGD_k)\}$ 
11 end

```

Studies in [3], [10], [41] point out the trade-off between the capacity of the local learning algorithm and the complexity. The large value of k reduces significant training time of k SGD and making a very low generalization capacity. The small value of k improves the generalization capacity but also increasing the training time.

To overcome the main memory limit of the Raspberry Pi, we propose to train local SGD models in the incremental fashion. The local training subset D tailored on the Raspberry Pi is split into T small blocks $\{D_1, D_2, \dots, D_T\}$. The incremental local SGD (denoted by Inc- k lSGD in Algorithm 3) sequentially loads data block D_t to learn local SGD models.

Finally, the federated learning algorithm collects via XML-RPC protocol vT local SGD models trained by Raspberry Pis to form the shared global classification model (denoted by FL- l SGD model = $\{kSGD_1, kSGD_2, \dots, kSGD_{vT}\}$). The federated learning algorithm of local SGD models can be explained by training an ensemble of local SGD models.

The prediction of a new datapoint x (in Algorithm 4) is the majority vote among classification results $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_{vT}$ obtained by vT k SGD models.

3 Experimental results

We are interested in the assessment of the federated learning algorithm of local SGD models (FL- l SGD) with the Raspberry Pis for classifying ImageNet challenging dataset. Therefore, it needs to evaluate the performance in terms of training time, demanded memory size and classification correctness.

Algorithm 2: Prediction of a new individual x with k SGD model

```

input :
    a new datapoint  $x$ 
     $k$ SGD-model =  $\{(c_1, SGD_1), (c_2, SGD_2), \dots, (c_k, SGD_k)\}$ 
output:
    predicted class  $\hat{y}$ 

1 begin
2   /* find the closest cluster based on the distance between  $x$  and cluster
   centers  $c_1, c_2, \dots, c_k$  */
3    $c_{NN} = \arg \min_c \text{distance}(x, c)$ 
4   /* the class of  $x$  is predicted by the local SGD model  $SGD_{NN}$ 
   corresponding to  $c_{NN}$  */
5    $\hat{y} = \text{predict}(x, SGD_{NN})$ 
6   return predicted class  $\hat{y}$ 
7 end

```

Algorithm 3: Incremental k local SGD (Inc- k SGD)

```

input :
    training dataset  $D = \{D_1, D_2, \dots, D_T\}$ 
    number of local models  $k$ 
    parameters of SGD  $\theta$ 
output:
    inc- $k$ SGD-model (ensemble of  $k$ SGD models)

1 begin
2   for  $t \leftarrow 1$  to  $T$  do
3     Loading data block  $D_t$ 
4      $kSGD_t = kSGD(D_t, k, \theta)$ 
5   end
6   return inc- $k$ SGD-model =  $\{kSGD_1, kSGD_2, \dots, kSGD_T\}$ 
7 end

```

Algorithm 4: Prediction of a new individual x with FL- l SGD model

```

input :
    a new datapoint  $x$ 
    FL- $l$ SGD model =  $\{kSGD_1, kSGD_2, \dots, kSGD_{vT}\}$ 
output:
    predicted class  $\hat{y}$ 

1 begin
2   for  $t \leftarrow 1$  to  $vT$  do
3      $\hat{y}_t = \text{predict}(x, kSGD_t)$ 
4   end
5    $\hat{y} = \text{majority-vote}\{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_{vT}\}$ 
6   return predicted class  $\hat{y}$ 
7 end

```

3.1 Software programs

We implemented the FL-*l*SGD in Python using library Scikit-learn [32]. The full SGD algorithm is already implemented in Scikit-learn. We would like to compare with the best state-of-the-art linear SVM algorithm, LIBLINEAR [14] implemented in C/C++ (the parallel version on multi-core computers with OpenMP [31]) and the full SGD algorithm.

Our FL-*l*SGD algorithm trains the ensemble of local SGD models with 4 Raspberry Pis (RPis) Raspbian Bullseyes, Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz, 4GB RAM. LIBLINEAR, full SGD learn classification models on a machine (PC) Linux Fedora 32, Intel(R) Core i7-4790 CPU, 3.6 GHz, 4 cores and 32 GB main memory.

3.2 ImageNet challenging dataset

Experimental results are evaluated on ImageNet challenging ILSVRC2012 dataset [5], [6] with 1,281,167 images and 1,000 classes which is the most popular visual classification benchmark [4], [5], [6], [7], [8], [9], [11], [12], [13], [17], [36], [38], [42].

We propose to use pre-trained EfficientNet [39] described in Fig. 3 to extract 1,280 invariant features from images (getting the last **GA pool** layer).

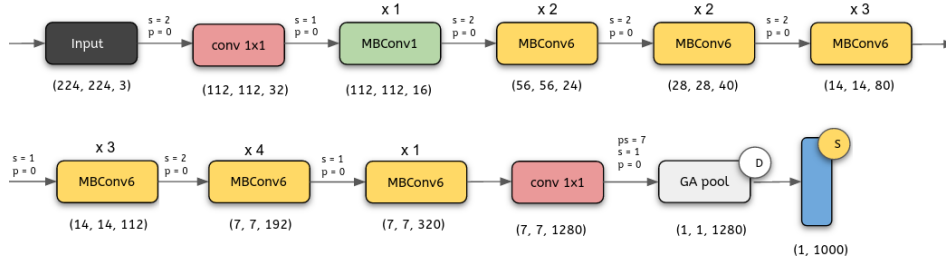


Fig. 3: Architecture of EfficientNet

ImageNet dataset is randomly divided into training set (1,024,933 images) and testing set (256,234 images) with random guess 0.1% due to 1,000 classes.

3.3 Tuning parameter

For training linear SVM models, it needs to tune the positive constant C in SVM algorithms for keeping the trade-off between the margin size and the errors. We use the cross-validation (hold-out) protocol to find-out the best value $C = 100,000$. LIBLINEAR uses L2-regularized Logistic Regression that is very closed to the softmax classifier used in deep learning networks, i.e. EfficientNet [39].

For the federated learning of local SGD models (FL-*l*SGD), the full training dataset is divided into 4 subsets which are stored in 4 Raspberry Pis. Therefore,

a Raspberry Pi contains 256,233 images. Furthermore, the local SGD algorithm at the Raspberry Pi trains local SGD models in the incremental fashion with the training block size being 128,116. The Raspberry Pi needs about 1GB RAM for learning local SGD models.

The parameter k local SGD models (number of clusters) of k SGD is set to 300 so that each cluster has about 500 datapoints. The idea gives a trade-off between the generalization capacity and the computational cost. Furthermore, the number of epochs and learning rate η of SGD are 50 and 0.0001, respectively.

Due to the Raspberry Pi 4 (Broadcom BCM2711, Quad core Cortex-A72) and the PC (Intel(R) Core i7-4790 CPU, 4 cores) used in the experimental setup, the number of threads is setting to 8 for all training tasks.

3.4 Classification results

We obtain classification results of FL- l SVM, k SVM and LIBLINEAR in Table 1, Fig. 4 and Fig. 5. The fastest training algorithm is in bold-faced and the second one is in italic. The same presentation format is accorded to performance in terms of classification accuracy, demanded memory size.

Given the differences in implementation, including the programming language (C++ versus Python), computer (PC Intel(R) Core i7-4790 CPU, 4 cores, 32 GB RAM versus Raspberry Pi 4 Broadcom BCM2711, Quad core Cortex-A72, 4GB RAM), the comparison of training time is not really fair. But our FL- l SGD algorithm achieves interesting results.

Table 1: Overall classification accuracy

No	Algorithm	Language	Machine	Required memsize (GB)	Time (min)	Accuracy (%)
1	FL- l SGD	Python	4 RPis	1	18.01	88.07
2	Full-SGD	Python	PC	<i>20</i>	<i>151.30</i>	88.81
3	LIBLINEAR	C/C++	PC	<i>20</i>	2,980.25	<i>88.21</i>

Our FL- l SGD with 4 Raspberry Pis classifies ImageNet dataset in 18.01 minutes with 88.07% accuracy. The full SGD using the PC achieves 88.81% accuracy with 151.30 minutes in the training time. LIBLINEAR with the PC takes 2,980.25 minutes for training the classification model with 88.21% accuracy.

In the comparison of training time among algorithms, we can see that the FL- l SGD with 4 Raspberry Pis is fastest training algorithm. Our FL- l SGD with 4 Raspberry Pis is 165.57, 4.68 times faster than LIBLINEAR and the full SGD with the PC, respectively. The full SGD is 19.70 times faster than LIBLINEAR.

In term of demanded memory size, the FL- l SGD training algorithm requires about 1 GB RAM against at least 20 GB RAM being used by the full SGD and LIBLINEAR.

In terms of overall accuracy, the full SGD achieves the highest accuracy of 88.81%. Followed which LIBLINEAR gives the accuracy of 88.21%. The FL- l SGD has the competitive classification correctness of 88.07%. The comparison,

algorithm by algorithm, shows that the superiority of LIBLINEAR and the full SGD on FL-*l*SGD corresponds to 0.14% and 0.74%, respectively.

The good performance of FL-*l*SGD algorithm is achieved by reducing the training complexity, the required main memory and parallelizing the training task of k local SGD models on multi-core computers.

The classification results show that our FL-*l*SGD algorithm is efficient for handling such large-scale multi-class datasets with Raspberry Pis.

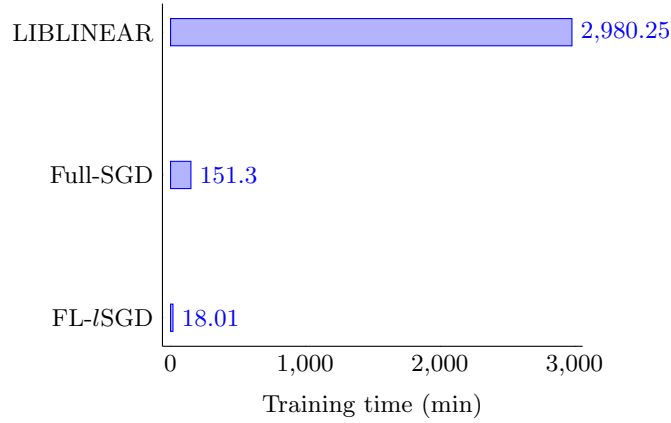


Fig. 4: Training time (min)

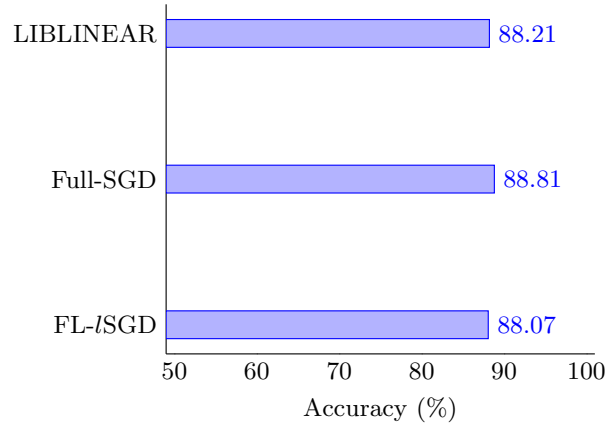


Fig. 5: Overall classification accuracy

4 Conclusion and future works

We have presented the new federated learning algorithm of local SGDs, called FL-*l*GD at the Raspberry Pi to deal with ImageNet challenge having 1,281,167 images with 1,000 classes. An ensemble of local SGD models in FL-*l*SGD are learned from local training subset at the Raspberry Pi without exchanging training data. Furthermore, the algorithm sequentially loads small data blocks of the local training subset to learn local SGD models. In which, it splits the data block into k partitions using k means algorithm and then it learns in the parallel way SGD models in each data partition to classify the data locally. The numerical test results on ImageNet challenging dataset show that our FL-*l*SGD algorithm with 4 Raspberry Pi is respectively 165.57 and 4.68 times faster than the state-of-the-art LIBLINEAR, the full SGD on the PC, with a competitive accuracy of 88.07%.

In the near future, we will develop the federated learning algorithms of local SGD models at various edge devices. In addition, we will study the weighted combination of local SGD models in the classification to improve the accuracy.

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